

PACH

Google Ads API Tool Design

Internal advertising workflow for Ardia Technologies S.A. de C.V. Prepared as supporting documentation for Google Ads API Basic Access.

Pach's initial publishing operation creates the complete Google Search campaign structure in **PAUSED** status. After API approval, activation, pause/resume, and controlled budget changes will be added as separate, explicitly approved actions.

COMPANY Ardia Technologies S.A. de C.V.	COMPANY URL https://ardia.mx
TOOL AND AUDIENCE Pach - internal employees and contractors only	MANAGER ACCOUNT Pach - customer ID 286-230-2046
BUSINESS TYPE Advertiser managing company-owned brands and business units	API ACCESS REQUESTED Basic Access for production advertiser accounts

Purpose of this document

This document describes Pach's current implementation and immediate product roadmap, including business context, users, Google Ads API operations, authentication, stored data, safety controls, error handling, and testing.

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Existing internal interface

Pach is an existing internal operations platform. The Promotions workspace imports eligible landing pages, generates keyword ideas, prepares Google Search drafts, and surfaces spend and performance status for review. The screenshot below is from the running Ardia organization workspace.

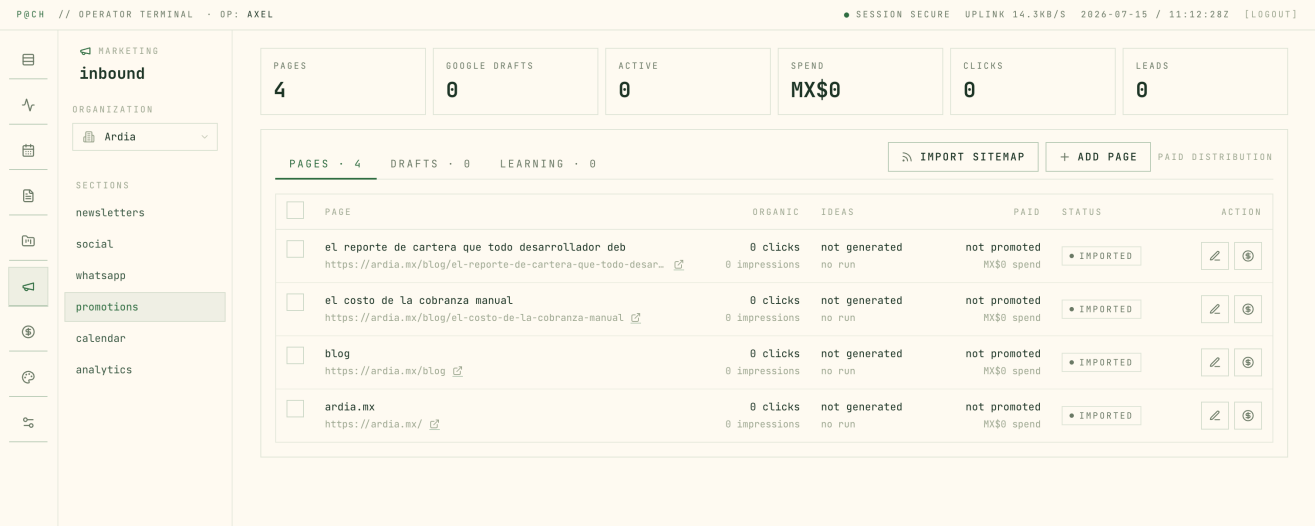


Figure 1. Pach Promotions workspace showing the Ardia organization, imported landing pages, Google draft count, spend, clicks, leads, and per-page review actions.

Visible area	Purpose
Organization selector	Scopes records and actions to the selected internal organization.
Pages	Lists HTTPS landing pages imported from the organization's sitemap or added manually.
Drafts	Holds local Google Search campaign drafts before any provider operation.
Paid status	Shows whether a page is promoted and summarizes spend and results when available.
Review actions	Opens page editing or campaign draft creation; no automatic publishing occurs.

02 / Business model and intended audience

Business model and intended audience

Ardia Technologies is a Mexican software company. Pach is developed and operated internally to coordinate marketing and operational activities for Ardia and company-owned brands or business units. Pach is not currently sold, licensed, or offered to the general public or unrelated external clients.

Authorized users

- Employees and approved contractors working for Ardia Technologies.
- Users must authenticate to Pach and must have access to the selected organization.
- A Google account with permission to the relevant manager or advertiser account must authorize OAuth access.
- Google advertiser accounts are selected per organization; manager accounts cannot be selected for publishing.

Primary use cases

Use case	Description	Human control
Account discovery	List accessible accounts and read name, ID, currency, timezone, type, test status, and provider status.	User synchronizes and selects one advertiser.
Draft preparation	Create local Search drafts from approved landing pages, keywords, copy, targeting, dates, and daily budget.	Draft remains local and editable.
Paused publishing	Create a budget, Search campaign, ad group, criteria, keywords, and responsive search ad.	Explicit confirmation; serving entities start PAUSED.
Campaign control	Activate, pause/resume, and update budget through separate API actions planned immediately after approval.	Explicit approval, account checks, limits, and audit trail.
Reporting	Read campaign status, impressions, clicks, cost, and conversions for Pach-linked campaigns.	Read-only scheduled sync and visible history.

Immediate roadmap after API approval

- Validate paused publishing against Ardia's production advertiser account.
- Add explicit activate and pause/resume actions with role checks, confirmation, and audit events.
- Add controlled daily-budget updates with configured caps and before/after values.
- Synchronize paid performance and provider campaign status for review and optimization.

Technical architecture and authentication

Pach uses a React portal, an authenticated Express/TypeScript API, and PostgreSQL. The browser never calls Google Ads directly. OAuth tokens and the developer token are handled only by the server.



OAuth 2.0 connection

- Pach uses its existing Google OAuth web client and requests <https://www.googleapis.com/auth/adwords>.
- The same authorized connection can retain Search Console scopes used by existing Pach functionality.
- Access and refresh tokens are encrypted before database storage and refreshed server-side.
- OAuth credentials are linked to an organization and the internal user who established the connection.

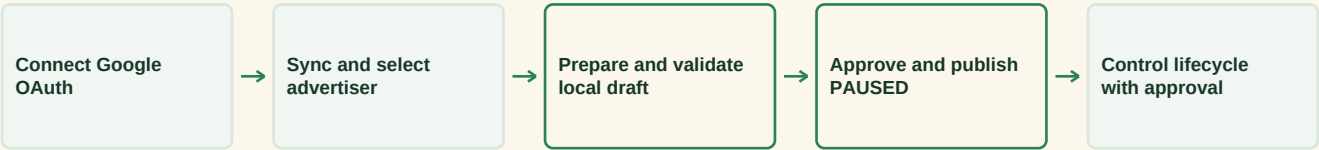
Google Ads request authentication

- The Authorization header uses the current OAuth bearer access token.
- developer-token: sourced from the `GOOGLE_ADS_DEVELOPER_TOKEN` server environment variable.
- login-customer-id: sent when the advertiser is accessed through the Pach manager account.
- The developer token is never returned to or stored in the browser.

API services and operations

Google Ads API operation	Purpose	Mutation
customers:listAccessibleCustomers	Discover manager and advertiser accounts visible to the authorized Google user.	No
GoogleAdsService.Search	Read account properties, manager children, target constants, and recover an existing idempotent campaign.	No
GoogleAdsService.Mutate	Create the full paused Search campaign structure in one atomic request.	Yes

Initial publishing and controlled campaign lifecycle



Step 1 - Connect and select

An authorized internal user connects Google, synchronizes accessible accounts, and selects a non-manager advertiser for the organization. Pach stores the customer ID, optional manager customer ID, currency, timezone, account type, test status, provider status, and last synchronization time.

Step 2 - Prepare and validate

The user prepares a local draft with a daily budget, HTTPS landing page, country and language, keywords, responsive search ad copy, and optional dates. Pach validates the budget, account currency, URL, field counts, and Google Ads text limits.

Step 3 - Explicit approval and atomic creation

A confirmation modal identifies the advertiser, budget, campaign, and landing page. After approval, Pach sends one `GoogleAdsService.Mutate` request with `partialFailure=false` and creates the following entities:

Entity	Initial configuration
Campaign budget	Positive daily amount, standard delivery, not shared.
Search campaign	Status PAUSED; Google Search only; selected country/language; configured dates.
Ad group	SEARCH_STANDARD; status PAUSED; initial CPC bid.
Keywords	Phrase match; each criterion created PAUSED.
Responsive search ad	Approved HTTPS URL; validated copy; ad status PAUSED.

Step 4 - Controlled lifecycle actions

The immediate next release will add separate activate, pause/resume, and budget-update operations. Each action will re-check organization and advertiser access, present the exact impact, require explicit confirmation, apply configured limits, and record the Google request and resulting state.

05 / Safety, lifecycle controls, and error handling

Safety, lifecycle controls, and error handling

Control	Implementation	Risk reduced
Paused first publish	Campaign, ad group, keyword criteria, and responsive search ad are initially created PAUSED.	Unreviewed delivery or spend.
Human confirmation	Every provider mutation is initiated through a specific confirmation action.	Accidental changes.
Advertiser guard	An active, non-manager advertiser must be selected for the same organization.	Wrong-account mutations.
Currency guard	Draft and budget currency must match the selected advertiser.	Incorrect budget interpretation.
Content validation	Requires HTTPS URL, keywords, minimum ad assets, and provider length limits.	Invalid or incomplete ads.
Staged activation	Activation and pause/resume are separate planned actions with fresh state checks and audit events.	Spend begins only after approval.
Controlled budgets	Planned updates show before/after values, enforce configured caps, and require approval.	Unexpected spend increases.
Atomic creation	Initial creation uses one mutate request with partialFailure=false.	Ambiguous partial campaign state.

Idempotency

- A promotion with an existing Google campaign ID returns the stored result without creating another campaign.
- A unique publish operation key claims the draft and prevents concurrent duplicate publishing.
- The campaign name includes the Pach promotion UUID; Pach searches the exact name and recovers IDs after an interrupted response.

Lifecycle state and concurrency

- Before activate, pause/resume, or budget changes, Pach will read or compare the latest provider state and reject stale or conflicting changes.
- High-impact actions will have one in-flight operation per campaign and will store the requested and resulting values.

Failure handling and auditability

- Google request IDs, resource names, API version, errors, and provider response details are stored for troubleshooting.
- Failed operations retain a clear local failed state and can be retried through the same guarded action.
- Activity events record account synchronization, publishing, activation state changes, and budget updates.

Data handling, testing, and release roadmap

Data stored by Pach

- Google connection identity, granted scopes, encrypted OAuth access/refresh tokens, status, and refresh timestamps.
- Advertiser metadata: customer IDs, name, currency, timezone, manager/test flags, selection, and last sync.
- Campaign configuration and outputs: landing page, budget, targeting, ad assets, Google resource IDs, status, request IDs, timestamps, and errors.
- Planned lifecycle audit records: prior and requested status or budget, resulting provider state, actor, and time.

Data not requested or stored

- No Google Account passwords, billing instruments, or payment credentials.
- No Customer Match lists, user identifiers, audience uploads, or offline conversion data.

Validation plan

Test	Expected result
OAuth without adwords scope	Account sync is blocked and reconnection is requested.
Manager or wrong advertiser	Mutation is blocked; the correct advertiser must be selected.
Valid initial draft	All campaign entities are created and remain PAUSED.
Activate or pause/resume	A confirmed action updates only the intended campaign and stores the provider result.
Budget update	Configured caps, currency, advertiser, and before/after values are validated before mutation.
Repeat or conflicting request	Idempotency/concurrency guard prevents duplicate or stale changes.
Provider/API error	Operation records request/error details and leaves an unambiguous retryable state.

Current release: account discovery, advertiser selection, local draft validation, and safe PAUSED publishing are implemented. Immediate post-approval release: production validation, activate, pause/resume, controlled budget updates, and paid performance synchronization.

Longer-term optimization

Once reporting and lifecycle controls are validated, Pach may recommend keyword, creative, targeting, and budget improvements. Material spend or delivery changes will remain bounded by configured policies, permissions, audit records, and human approval until separate automation controls are proven.